Mohamad **Dikshie Fauzie**

Forward Deployed Engineer · Applied AI, Agentic Systems & Cloud Solutions

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Summary

Senior Systems Engineer and Applied AI Solutions Architect with 10+ years of mission-critical production experience bridging enterprise customer requirements with production-ready software architectures and AI-driven solutions. Holds a Ph.D. in Distributed Systems from Keio University (supervised by Internet Hall of Fame founder Prof. Jun Murai) and a B.Sc. in Physics from ITB. Proven track record in forward-deployed technical delivery: engaging directly with enterprise clients and financial participants, translating ambiguous business challenges into scalable application workflows, and rapidly driving prototypes into resilient production systems. Combines hands-on Python/API engineering, Generative AI application development, enterprise RAG pipelines (chunking, hybrid retrieval, reranking), and agent orchestration with full-lifecycle MLOps/LLMOps observability (Prometheus, Grafana). Certified in NVIDIA AI Infrastructure, AWS Cloud, and Google Project Management. Tokyo resident with business Japanese (JLPT N2 level) and fluent English, fully prepared for high-impact on-site customer engagement.

Skills

Applied AI & GenAI	Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration (LangChain, LlamaIndex), Prompt Engineering, Semantic Search, Vector Databases (FAISS, Qdrant), Model Evaluation & Benchmarking
Application Eng.	Python (FastAPI, Flask, AsyncIO, PyTest), JavaScript/Node.js, RESTful APIs, Microservices Architecture, Event-Driven Systems, Modular Design, Git CI/CD, Containerization (Docker)
Cloud & LLMOps	AWS (Certified), Kubernetes, Linux (RHEL, Ubuntu), MLOps/LLMOps Lifecycle (Deployment, Versioning, Monitoring, Rollback), AI Observability (Prometheus, Grafana, OpenTelemetry), Logging & Metrics
Systems & Data	Distributed Systems, Dell PowerScale (Isilon) Parallel Storage, PostgreSQL, Redis, NetBox (DCIM), High-Throughput Telemetry Ingestion, Data Security, Access Control (RBAC), Audit Governance
Client Delivery	Forward Deployed Engineering, Customer-Facing Solutions Architecture, Requirements Translation, Rapid Prototyping to Production, Stakeholder Alignment, Google Certified Project Management
Languages	English (Fluent / Business Required), Japanese (Business Proficiency / JLPT N2 level, 15+ yrs in Tokyo), Indonesian (Native)

Work Experience

Japannext Co., Ltd. (Largest Proprietary Trading System / PTS in Japan)

Tokyo, Japan

Forward Deployed Systems Architect & Technical Delivery Lead (Co-Manager) — *Oct. 2014 - Present*

- Served as primary customer-facing technical lead for institutional participants, broker-dealers, and technology partners, translating complex connectivity and data processing requirements into production-ready technical architectures.
- Architected and deployed Python-based microservices, REST APIs, and automated telemetry ingestion pipelines, eliminating manual operational reporting and achieving 100% automated SLA compliance reporting.
- Built operational telemetry and incident trend analytics platforms, analyzing high-volume event logs and ticket patterns to optimize engineering resource allocation, accelerate MTTR, and achieve 50% power distribution balancing across high-density systems.
- Designed and deployed open-source DCIM inventory tracking (NetBox) and real-time observability pipelines (Prometheus, Grafana), accelerating fault isolation and providing deep operational visibility across multi-datacenter environments (TY3, CC2).
- Spearheaded end-to-end datacenter migrations under tight power and operational constraints; executed seamless cutovers with zero downtime 2 weeks ahead of schedule, unlocking 50% system expansion capacity.
- Architected high-throughput parallel data pipelines on Dell PowerScale (Isilon) storage and automated LTO backup workflows, gaining 50% I/O throughput for multi-terabyte daily data logs while maintaining strict FSA regulatory data governance and security controls.

Keio University - JICA JST Indo Project

Tokyo, Japan

Research Assistant (Distributed Systems, Algorithms & AI) — *Apr. 2011 - Sep. 2014*

- Researched distributed network topologies, peer-assisted CDNs, and probabilistic gossip protocols, modeling complex network dynamics and information dissemination under node churn.
- Engineered high-performance simulation models in Python and C++, evaluating graph dynamics and energy optimization; published 3 peer-reviewed papers in IEEE and IEICE conferences and journals.

Keio University

Tokyo, Japan

System Administrator (Campus Core Infrastructure) — *Oct. 2007 - Sep. 2014*

- Administered campus-wide core infrastructure (BGP/OSPF routing, DNS, email, directory services, and firewall security), ensuring 99.99% uptime for research laboratories and academic faculty.

Applied AI & Engineering Projects

Enterprise Retrieval-Augmented Generation (RAG) Intelligence Pipeline

Python, LangChain/LlamaIndex, Qdrant, Docker

Lead Architect & Developer — *2024*

- Designed and deployed an end-to-end RAG system enabling semantic search and conversational question-answering over complex technical manuals, API documentation, and regulatory operational logs.
- Engineered modular document ingestion pipelines featuring semantic chunking, dense vector embeddings, hybrid retrieval (BM25 + dense vector search), and cross-encoder reranking to optimize context recall and precision.
- Implemented automated evaluation frameworks measuring answer faithfulness, context relevance, and hallucination rates, establishing repeatable benchmarks for model quality assurance.

Autonomous Multi-Agent Task Orchestration & Diagnostic Workflow

Python, FastAPI, OpenAI/Anthropic APIs, Pydantic

Architect & Developer — *2023 - 2024*

- Engineered a modular agent-based architecture with dynamic tool integration to automate complex diagnostic query workflows and multi-step data retrieval tasks.
- Implemented robust task orchestration logic, structured JSON schema validation, error recovery loops, and human-in-the-loop escalation gates to guarantee execution reliability in production workflows.

LLMOps Observability & Telemetry Pipeline

Python, Prometheus, Grafana, OpenTelemetry, Docker

Architect & Developer — *2023 - 2024*

- Built an end-to-end LLMOps telemetry collector tracking token consumption, inference latency distributions, cost attribution, and prompt failure rates across production model invocations.
- Designed consolidated operational Grafana dashboards with automated alerting to monitor model drift and ensure SLA reliability.

Education

Keio University

Tokyo, Japan

Ph.D. in Media and Governance (Internet Architecture & Distributed Systems) — *Oct. 2014*

- Dissertation: Dynamic P2P network topologies, swarm dynamics, and energy-efficient Content Delivery Networks (CDNs).
- Supervisors: Prof. Jun Murai (Internet Hall of Fame / WIDE Project), Prof. Rodney Van Meter (Quantum Networking Pioneer).

Institut Teknologi Bandung (ITB)

Bandung, Indonesia

M.T. (Master of Technology in Electrical Engineering) — *Oct. 2004*

- Designed and benchmarked one of Indonesia's earliest IPv6 campus network backbones and core routing deployments.

Institut Teknologi Bandung (ITB)

Bandung, Indonesia

S.Si. (Bachelor of Science in Physics) — *Oct. 2000*

- Researched data assimilation and numerical simulation in 1D hydrodynamic fluid systems (differential wave equations).

Certifications & Specialized Training

2024	**AI Infrastructure and Operations Fundamentals**, NVIDIA (Deep Learning Institute)	*Online*
2024	**Foundations of Project Management**, Google Career Certificates	*Online*
2023	**Generative AI for Everyone**, DeepLearning.AI	*Online*
2023	**Mathematics for Machine Learning and Data Science**, DeepLearning.AI	*Online*
2023	**AWS Certified Cloud Practitioner**, Amazon Web Services (AWS)	*Online*
2023	**Machine Learning**, Stanford University	*Online*
2019	**Cloud Infrastructure, Kubernetes & OpenStack**, The Linux Foundation	*Online*

Publications

Energy Consumption in Peer Assisted CDN

International Journal of Advanced Research in Computer Science (IJARCS) Vol.4 No.9

Lead Author — *Aug. 2013*

- Modeled and evaluated energy efficiency of peer-assisted edge CDN architectures versus traditional centralized distribution.

Assessing The Dynamics of BitTorrent Swarm Topologies Using The Peer Exchange Protocol

IEICE Transactions on Communications (Special Issue on Information Network Science, Vol.E95-B, No.05)

Lead Author — *2012*

- Empirically inferred dynamic overlay topologies and peer discovery mechanisms using the Peer Exchange (PEX) protocol.

A Temporal View of The Topology of Dynamic BitTorrent Swarms

IEEE INFOCOM Workshop on Network Science for Communication Networks

Lead Author — *2011*

- Analyzed temporal graph evolution and topological robustness of large-scale distributed swarms under dynamic churn.